ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 29	$(\alpha+\beta)^2-2\alpha\beta=7^2-2(10)$
2	**(d)** $\frac{5}{6}$	$\dfrac{\alpha+\beta}{\alpha\beta}$
3	**(c)** 10	$2^2-7(2)+k=0$
4	**(a)** 3	$2^2+2k-10=0$
5	**(c)** -7	$p=-(\alpha+\beta)=-(1+6)$
6	**(d)** 4	Sum of zeroes $=7$, so other zero $=7-3$
7	**(c)** 25	$(\alpha+\beta)^2-2\alpha\beta=7^2-2(12)$
8	**(a)** $\frac{5}{4}$	$\dfrac{\alpha+\beta}{\alpha\beta}$
9	**(b)** 15	$3^2-8(3)+k=0$
10	**(d)** -1	$3^2+3k-6=0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\alpha+\beta=8$, $\alpha\beta=12$ (1); $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=40$ (1).	1+1
12	$p(3)=0$: $-15+k=0$ (1) so $k=15$ (1).	1+1
13	New sum $=2(3)=6$, product $=4(2)=8$ (1); polynomial x^2-6x+8 (1).	1+1
14	$\alpha+\beta=10$, $\alpha\beta=21$ (1); $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=58$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha+\beta=6$, $\alpha\beta=8$ (1); $\dfrac{\alpha+\beta}{\alpha\beta}=\frac{3}{4}$ (1); $\alpha^2+\beta^2=20$ (1).	1+1+1
16	Sum of new zeroes $=\dfrac{\alpha+\beta}{\alpha\beta}=\frac{5}{6}$ (1); product $=\dfrac{1}{\alpha\beta}=\frac{1}{6}$ (1); polynomial $x^2-(\frac{5}{6})x+(\frac{1}{6})$ (1).	1+1+1
17	$p(6)=0$: $36-66+30=0$ gives $k=-11$ (1); sum of zeroes $=-k=11$ (1); other zero $=5$ (1).	1+1+1
	OR: $\alpha+\beta=6$, $\alpha\beta=5$ (1); $\dfrac{\alpha+\beta}{\alpha\beta}=\frac{6}{5}$ (1); $\alpha^2+\beta^2=26$ (1).	1+1+1
18	Sum of new zeroes $=\dfrac{\alpha+\beta}{\alpha\beta}=\frac{7}{6}$ (1); product $=\dfrac{1}{\alpha\beta}=\frac{1}{6}$ (1); polynomial $x^2-(\frac{7}{6})x+(\frac{1}{6})$ (1).	1+1+1
	OR: $p(-5)=0$: $25-15-10=0$ gives $k=3$ (1); sum of zeroes $=-k=-3$ (1); other zero $=2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of the three zeroes $=-\dfrac{b}{a}=6$ (2); third zero $=6-(1)-(2)=3$ (2); zeroes $1, 2, 3$ (1).	2+2+1
	OR: Sum of zeroes $=-\dfrac{b}{a}=\dfrac{12}{k}=3$ (2); $k=4$ (1); product $=\dfrac{c}{a}=\dfrac{8}{4}=2$ (2).	2+1+2
20	Sum of the three zeroes $=-\dfrac{b}{a}=6$ (2); third zero $=6-(-1)-(2)=5$ (2); zeroes $-1, 2, 5$ (1).	2+2+1
	OR: Sum of zeroes $=-\dfrac{b}{a}=\dfrac{15}{k}=5$ (2); $k=3$ (1); product $=\dfrac{c}{a}=\dfrac{9}{3}=3$ (2).	2+1+2

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $=-\frac{b}{a}$ and product $=\frac{c}{a}$.
- Forgetting the sign: for $x^2-8x+15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y=p(x)$ as the y-intercept; zeros are where the graph meets the x-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.

- Writing the polynomial from its zeros as $x^2 + (\alpha + \beta)x + \alpha\beta$ instead of $x^2 - (\alpha + \beta)x + \alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** -9	$p = -(\alpha+\beta) = -(2+7)$
2	**(b)** 5	Sum of zeroes $= 9$, so other zero $= 9-4$
3	**(a)** 37	$(\alpha+\beta)^2 - 2\alpha\beta = 5^2 - 2(-6)$
4	**(a)** $\frac{8}{15}$	$\frac{\alpha+\beta}{\alpha\beta}$
5	**(c)** 20	$4^2 - 9(4) + k = 0$
6	**(c)** -2	$-2^2 + -2k - 8 = 0$
7	**(a)** -8	$p = -(\alpha+\beta) = -(3+5)$
8	**(a)** 5	Sum of zeroes $= 8$, so other zero $= 8-3$
9	**(c)** 65	$(\alpha+\beta)^2 - 2\alpha\beta = 9^2 - 2(8)$
10	**(b)** $\frac{2}{3}$	$\frac{\alpha+\beta}{\alpha\beta}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p(2) = 0$: $-14 + k = 0$ (1) so $k = 14$ (1).	1+1
12	New sum $= 2(5) = 10$, product $= 4(6) = 24$ (1); polynomial $x^2 - 10x + 24$ (1).	1+1
13	$\alpha+\beta = 3$, $\alpha\beta = -10$ (1); $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 29$ (1).	1+1
14	$p(4) = 0$: $-12 + k = 0$ (1) so $k = 12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha+\beta = 9$, $\alpha\beta = 18$ (1); $\frac{\alpha+\beta}{\alpha\beta} = \frac{1}{2}$ (1); $\alpha^2+\beta^2 = 45$ (1).	1+1+1
16	Sum of new zeroes $= \frac{\alpha+\beta}{\alpha\beta} = \frac{7}{12}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{12}$ (1); polynomial $x^2 - (\frac{7}{12})x + (\frac{1}{12})$ (1).	1+1+1
17	$p(7) = 0$: $49 - 70 + 21 = 0$ gives $k = -10$ (1); sum of zeroes $= -k = 10$ (1); other zero $= 3$ (1).	1+1+1
	OR: $\alpha+\beta = 5$, $\alpha\beta = -14$ (1); $\frac{\alpha+\beta}{\alpha\beta} = -\frac{5}{14}$ (1); $\alpha^2+\beta^2 = 53$ (1).	1+1+1
18	Sum of new zeroes $= \frac{\alpha+\beta}{\alpha\beta} = \frac{7}{10}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{10}$ (1); polynomial $x^2 - (\frac{7}{10})x + (\frac{1}{10})$ (1).	1+1+1
	OR: $p(-4) = 0$: $16 - 36 + 20 = 0$ gives $k = 9$ (1); sum of zeroes $= -k = -9$ (1); other zero $= -5$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of the three zeroes $= -\frac{b}{a} = 1$ (2); third zero $= 1 - (2) - (3) = -4$ (2); zeroes $2, 3, -4$ (1).	2+2+1
	OR: Sum of zeroes $= -\frac{b}{a} = \frac{14}{k} = 2$ (2); $k = 7$ (1); product $= \frac{c}{a} = \frac{21}{7} = 3$ (2).	2+1+2
20	Sum of the three zeroes $= -\frac{b}{a} = 5$ (2); third zero $= 5 - (1) - (-2) = 6$ (2); zeroes $1, -2, 6$ (1).	2+2+1
	OR: Sum of zeroes $= -\frac{b}{a} = \frac{18}{k} = 6$ (2); $k = 3$ (1); product $= \frac{c}{a} = \frac{12}{3} = 4$ (2).	2+1+2

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $= -\frac{b}{a}$ and product $= \frac{c}{a}$.
- Forgetting the sign: for $x^2 - 8x + 15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y = p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.

- Writing the polynomial from its zeros as $x^2 + (\alpha + \beta)x + \alpha\beta$ instead of $x^2 - (\alpha + \beta)x + \alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 30	$5^2 - 11(5) + k = 0$
2	**(d)** -2	$5^2 + 5k - 15 = 0$
3	**(b)** -10	$p = -(\alpha + \beta) = -(2 + 8)$
4	**(d)** 7	Sum of zeroes $= 10$, so other zero $= 10 - 3$
5	**(a)** 34	$(\alpha + \beta)^2 - 2\alpha\beta = 2^2 - 2(-15)$
6	**(d)** $\frac{9}{20}$	$\frac{\alpha + \beta}{\alpha\beta}$
7	**(d)** 24	$6^2 - 10(6) + k = 0$
8	**(a)** -1	$-3^2 + -3k - 12 = 0$
9	**(b)** -13	$p = -(\alpha + \beta) = -(4 + 9)$
10	**(c)** 8	Sum of zeroes $= 11$, so other zero $= 11 - 3$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New sum $= 2(3) = 6$, product $= 4(-4) = -16$ (1); polynomial $x^2 - 6x - 16$ (1).	1+1
12	$\alpha + \beta = 10$, $\alpha\beta = 9$ (1); $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 82$ (1).	1+1
13	$p(5) = 0$: $-5 + k = 0$ (1) so $k = 5$ (1).	1+1
14	New sum $= 2(8) = 16$, product $= 4(15) = 60$ (1); polynomial $x^2 - 16x + 60$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha + \beta = 11$, $\alpha\beta = 18$ (1); $\frac{\alpha + \beta}{\alpha\beta} = \frac{11}{18}$ (1); $\alpha^2 + \beta^2 = 85$ (1).	1+1+1
16	Sum of new zeroes $= \frac{\alpha + \beta}{\alpha\beta} = \frac{5}{12}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{24}$ (1); polynomial $x^2 - (\frac{5}{12})x + (\frac{1}{24})$ (1).	1+1+1
17	$p(8) = 0$: $64 - 112 + 48 = 0$ gives $k = -14$ (1); sum of zeroes $= -k = 14$ (1); other zero $= 6$ (1).	1+1+1
	OR: $\alpha + \beta = 12$, $\alpha\beta = 32$ (1); $\frac{\alpha + \beta}{\alpha\beta} = \frac{3}{8}$ (1); $\alpha^2 + \beta^2 = 80$ (1).	1+1+1
18	Sum of new zeroes $= \frac{\alpha + \beta}{\alpha\beta} = \frac{9}{8}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{8}$ (1); polynomial $x^2 - (\frac{9}{8})x + (\frac{1}{8})$ (1).	1+1+1
	OR: $p(-7) = 0$: $49 - 28 - 21 = 0$ gives $k = 4$ (1); sum of zeroes $= -k = -4$ (1); other zero $= 3$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of the three zeroes $= -\frac{b}{a} = 6$ (2); third zero $= 6 - (3) - (4) = -1$ (2); zeroes $3, 4, -1$ (1).	2+2+1
	OR: Sum of zeroes $= -\frac{b}{a} = \frac{20}{k} = 4$ (2); $k = 5$ (1); product $= \frac{c}{a} = \frac{15}{5} = 3$ (2).	2+1+2
20	Sum of the three zeroes $= -\frac{b}{a} = 4$ (2); third zero $= 4 - (2) - (5) = -3$ (2); zeroes $2, 5, -3$ (1).	2+2+1
	OR: Sum of zeroes $= -\frac{b}{a} = \frac{16}{k} = 8$ (2); $k = 2$ (1); product $= \frac{c}{a} = \frac{6}{2} = 3$ (2).	2+1+2

Mistakes to avoid:

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $= -\frac{b}{a}$ and product $= \frac{c}{a}$.
- Forgetting the sign: for $x^2 - 8x + 15$ the sum of zeros is $+8$, not -8.
- Counting the zeros of $y = p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
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